

Entropy Generation

Is it energy lost?

Energy quantity vs. energy quality, with worked examples

Not energy lost. Work potential lost.

Energy is always conserved; its ability to do work is not

First Law

Quantity

Energy is conserved. Friction, heat transfer, mixing: every joule is still there afterwards.

Unit: J

Second Law

Quality

Energy quality is always being destroyed.
Entropy generation measures that destruction.

Unit: J/K

Gouy–Stodola theorem: $W_{\text{lost}} = T_0 \times S_{\text{gen}}$

Multiply entropy generated by ambient temperature to get the useful work destroyed.

Where entropy is generated

Energy conserved in every case, yet work potential destroyed

1

Heat transfer across ΔT

Hot side loses Q/T_H , cold side gains Q/T_C
 $> Q/T_H$. Larger ΔT , more S_{gen} .

2

Friction & fluid losses

Pipe pressure drop becomes internal energy.
Pump work ends up as entropy.

3

Throttling

Valve: h constant, p drops, s rises. Expansion
that could have driven a turbine is wasted.

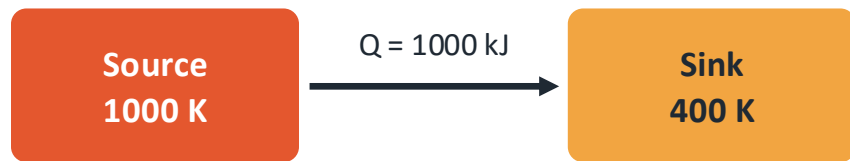
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Mixing

Hot + cold water, or two gases. Energy
conserved, but it never unmixes by itself.

Worked example: heat across a large ΔT

1000 kJ flows from a 1000 K source to a 400 K sink; ambient $T_o = 300$ K



Entropy balance

$$S_{\text{gen}} = Q/T_C - Q/T_H$$
$$= 1000/400 - 1000/1000 = 1.5 \text{ kJ/K}$$

Carnot check (rejecting to 300 K)

$$\text{At } 1000 \text{ K: } 1000 \times (1 - 300/1000) = 700 \text{ kJ}$$

$$\text{At } 400 \text{ K: } 1000 \times (1 - 300/400) = 250 \text{ kJ}$$

0 kJ

Energy lost

1.5 kJ/K

Entropy generated

450 kJ

Work destroyed = 300×1.5

Power plant: where is the real loss?

Coal-fired steam plant: energy analysis vs. exergy analysis

Condenser

Largest ENERGY rejection

- Rejects the most heat to cooling water
- But that heat is already near ambient temperature
- Low quality: little work potential left to lose

Boiler

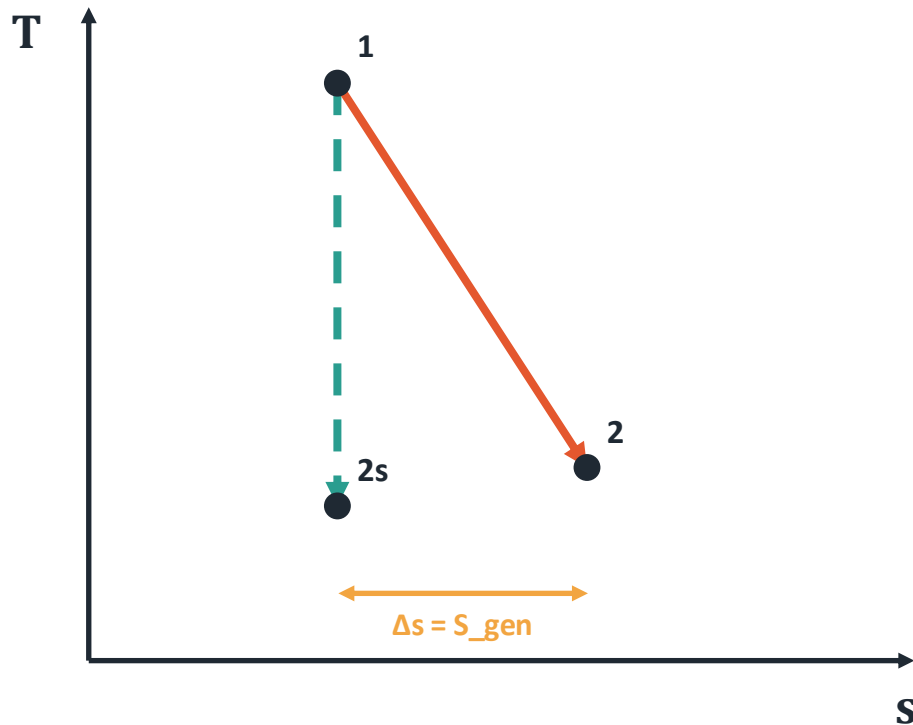
Largest EXERGY destruction

- Heat from ~ 1800 K flue gas to ~ 800 K steam
- Huge ΔT means huge entropy generation
- Most work potential in the fuel is destroyed here

Takeaway: to improve efficiency, target where entropy is generated, not just where energy leaves.

Seeing it on a T-s diagram

Turbine expansion: ideal (isentropic) vs. real



● Ideal (1→2s)

Vertical line: adiabatic and reversible. No entropy generated, maximum work.

● Real (1→2)

Slants right: friction and turbulence generate entropy. Less work out.

● The gap

Horizontal shift is S_{gen} . Multiply by T_0 for work lost; this is why we use isentropic efficiency.

Key takeaways

1

Energy is never lost

First law: quantity is always conserved.

2

Quality is always destroyed

Second law: $S_{\text{gen}} \geq 0$ in every real process.

3

S_{gen} is not energy

J/K, not J. Lost work = $T_0 \times S_{\text{gen}}$.

4

Design to minimise S_{gen}

Smaller ΔT s, less friction, fewer throttles, less mixing.